Italicized text means replace it with your text.

SSC Field Measurement Plan

Revised: *(insert date of revision here)*

Edmonds Test Deployment for Puget Sound

Mission start and end date here – Apr 18, 2025

POC: Kathryn Farabaugh (kfara@uw.edu, 302-373-6332)

Overview *(~1 paragraph)*

The main goal of this mission is to provide the science and tech team an opportunity to practice deploying out of Edmonds to prepare for the Puget Sound Time Series deployment later in the quarter. The SSC has not previously deployed out of Edmonds so this deployment will ensure pilots are familiar with the bathymetry of location and finalize targets for the puget sound time series. Additionally, this deployment will be used to train new SSC members on mission planning, deployment, piloting, and recovery best practices. For this mission glider sg175 will be deployed. This document will describe the technical and logistical details of the intended data collection.

Location *(~1 paragraph + relevant picture (i.e. maps with markers of waypoints)*

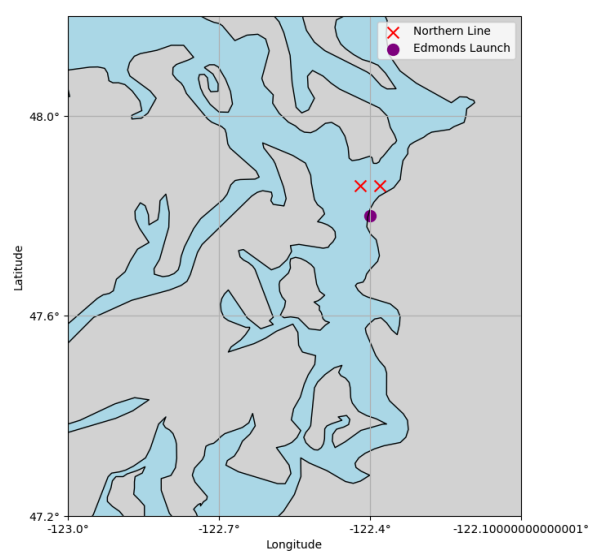
Describe the location of the mission. How remote is the location? Are there cultural differences that should be noted (i.e. working with locals to achieve smooth and safe work)? Is there a POC (i.e. chief scientist) for this mission? How do people get to this location (plane? Ship? etc.)? Describe some brief stats about the location (average depth, currents, winds, etc).

Boat Launch: Edmonds Boat Launch

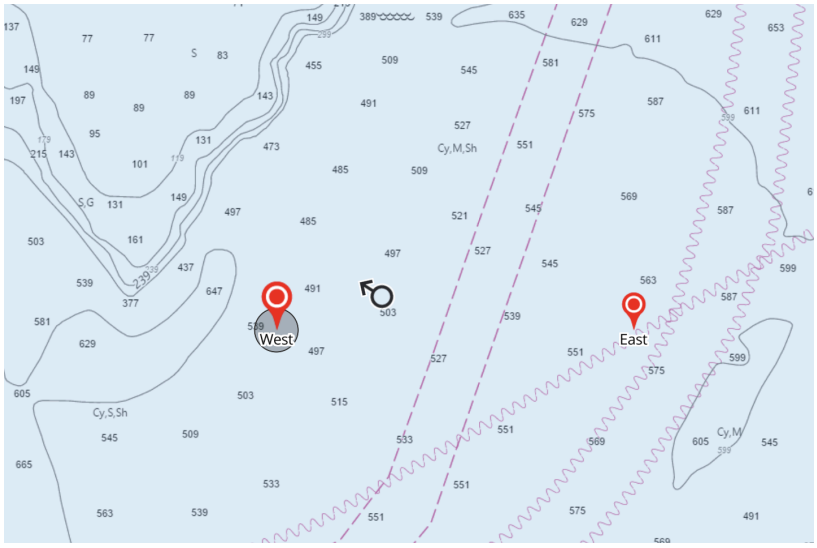
Tentative Waypoints (Degrees, Minutes, Secs)

Launch Point East: 47°51'36"N 122°22'24"W

West: 47°51'36"N 122°24'24"W

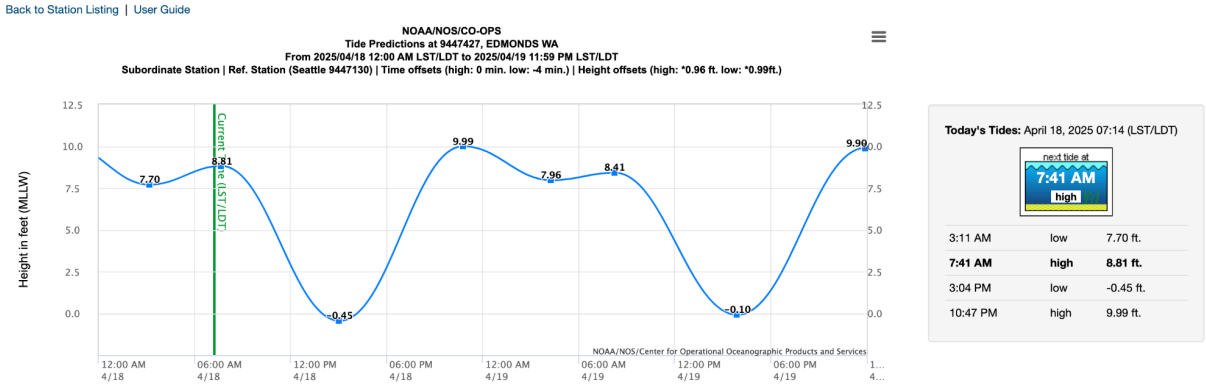


Bathymetry Map:



- Max depth in region ~ 167 m
- Min depth in region ~ 149 m
- North West of our way points, the depth of the water becomes much more shallow. Important to remember if the glider starts to drift.

Tide Chart April 18 2025 Edmonds



Main Science Questions *(~1-2 paragraphs + figures as needed)*

This is a test deployment and is not a scientific mission.

Goals *(~2 paragraphs + figures as needed)*

The goal of this mission is to prepare for the upcoming Puget Sound Time Series. This deployment will train new pilots and tech team members.

To train new members we hope to complete 45, 60m, max depth, and compass calibration dives. Below is the breakdown of the dives we

Type	Goal number of dives	Other information	Time estimate (T_MISSION)
45 m	1	We will start with this shallow dive to ensure glider is flying properly	20 min each
60m	1-2		30 min each
Max depth dives (120 m)	2		60 min each
Compass calibration dives	2	Compass calib guide doc MAKE SURE TO COPY DOWN REG PARAMETERS **BEFORE** PUTTING IN COMPASS CAL	70 min each

Over all CONSERVATIVE time amount needed to achieve goals: ~6 hours (7 dives)

Logistics *(~3 paragraphs)*

All communication will occur in "test-deployment" channel

Pilot POC: Kathryn

Field logistics:
Field POC: Christian
Field Team for 4/18:

- Christian
- Ziggy
- Sage

Field schedule
6:00 am: Arrival at OSB
6:50 am: Depart to Edmonds
7:30 am: Arrive at Edmonds Boat Launch, boat setup/run self-test (~20 minutes)
7:50 am: Depart to deployment coordinates (47°51'36"N 122°22'24"W)
8:30 am: Deploy Glider
8:35 - 9:30 am: Watch first dive, return to Edmonds Boat Launch afterwards
1:50 pm: Transit to Glider pick up
2:30 - 3:00 pm: Glider pick up
4:00 pm: Return to UW
4:40 pm: ETA at OSB

Mobile Instruments *(~3 paragraphs + pictures/diagrams)*

What glider(s) will be deployed? What sensors will they have on them and what will they be measuring? How will they be deployed? How will the data be transmitted/stored? Who will be piloting them and how?

Describe the dives here. How deep will they go? What do their paths look like (put a map of their waypoint here along with their tracks)? What time of day will they be going in and why (i.e. tide considerations).

Put a table of glider piloting parameters that will be specified for the base station here. This should be enough information for anybody to set up the glider for the mission. Take pictures of the cmdfile, targets file, and science files and upload them here.

- Glider 175 will be deployed
- Waypoints are east to west
- 45m, 60m, and max depth dives. (3-5 dives)
- compass calib dives (2 dives)

Glider Constants and Ballasting *(~1 paragraph + diagrams+figures as needed)*

List any relevant information on the planned area here. Include surface and dive depth densities, typical tidal currents, and any information that will be relevant for the tech team to adjust the gliders with.

Glider Details ***(Table)***

Describe all of the components and dimensions in a table for each glider. Adjust as needed.

Dimensions	
Power	
Hull material	
Weight	
Shipping crate dimensions	
Endurance	
Tracking (RF)	
Tracking (Iridium)	
Telemetry	
Processor	
Sensors…	

Instrumentation Summary *(table)*

Describe the instruments on the glider in this table (add or remove rows/columns as needed). Look these up on the sensor's spec sheets and report anything that is important (the more, the better).

Instrument	**Platform**	**Sampling Rate**	**Precision**	**Sampling Depth**	**Resolution**	**Power draw**	**Last calibration date**

Seabird 911plus CTD	*SSC SG195*	*1 Hz*	*0.01 deg C*	*0-1000 m*	*3 m bins*		

Testing *(~1 paragraph)*

Describe the relevant tests needed before this deployment here. Where will the self test occur and what is needed to get the glider(s) ready for the mission.

Personnel *(table)*

Who will be conducting all of the mission steps listed above? Arrival and departure are for people who will be going to the actual mission site. Include everyone involved in testing, ballasting, and piloting and such here.

Name	**Email**	**Team**	**Arrival IGG**	**Departure IGG**
Kathryn Farabaugh	kfara@uw.edu	Science Team		

Budget *(table)*

Describe the budget for everything included in the mission. This includes sensors, ship time, travel and lodging costs, and everything else. Include total at the bottom.

Action item/sensor	**Cost**

Safety *(~3 paragraphs)*

Describe the overall safety requirements/expectations for this mission.

List small boat operations and communications if needed.

Describe incident response here.

Appendix

List any supporting information here. This includes plots, maps, calculations, and any relevant information that can help describe this mission.

References

List references from any papers/websites here